

## John A. Scarrow

Coeur d'Alene, ID

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## Summary

Aspiring software engineer with two years of focused academic training in **C++**, **Python**, and **computer science** at North Idaho College, combined with a **1-year Full-Stack JavaScript program** (Code Fellows). Strong foundation in backend development, algorithms, cloud deployment, and system-level problem solving. Experienced with **Docker**, **AWS**, **GitHub workflows**, and AI-assisted development tools. Eager to contribute to modern .NET-based systems and learn **ASP.NET Core**, **Razor Pages**, and **Entity Framework Core**.

## Technical Skills

- **Languages:** C++, Python, JavaScript, SQL, JSON, LC-3 Assembly
- **Web / Backend:** REST APIs, Express.js, Node.js, Flask
- **Cloud / DevOps:** Docker, AWS S3, GitHub, Heroku, VM deployment
- **Tools:** SFML, React.js, MongoDB, MySQL, OAuth2, AJAX
- **Other:** Computer-vision annotation (CVAT), ML training pipelines, Windows Server configuration

## Relevant Coursework (North Idaho College)

- **C++ Programming I & II (2 years):** OOP, memory management, data structures, algorithm design
- **Python Programming (2 years):** scripting, automation, data processing
- **Computer Science Foundations:** logic, algorithms, computational thinking
- **Introduction to Robotics:** sensors, actuators, microcontrollers

## **Selected Projects** (Project titles are links)

### **[Battleship-ML \(C++ + ML\)](#)**

- Built a modular C++ game engine with reinforcement-learning and heuristic AI opponents
- Implemented a Python-based ML training pipeline
- Deployed a WebAssembly version for browser-based interaction

### **Python RESTful API with Docker & AWS**

- Developed production-ready REST APIs using Flask and MongoDB
- Containerized services with Docker and deployed using AWS DevOps practices

### **[Computer Vision Annotation \(CVAT.ai\)](#)**

- Annotated bounding boxes, skeletons, and interpolation for robotics/AI model training
- Gained practical experience with ML-ready datasets

### **[Bounty of Juan \(C++17 + SFML\)](#)**

- Designed wave-based enemy logic and optimized rendering performance
- Demonstrated strong C++ architecture and real-time systems design

### **[WildWest Shootout — High Noon \(LC-3 Assembly\)](#)**

- Built custom TRAP routines and data structures
- Strengthened understanding of low-level systems and CPU architecture

# Experience

## Freelance Developer — Coeur d'Alene, ID

*March 2020 – Present*

- Designed and maintained full-stack web applications
- Implemented backend features, debugging, and iterative improvements
- Collaborated with clients to deliver stable, maintainable solutions

## Snoqualmie Tribe Market — Snoqualmie, WA

*Supervisor, POS Support | March 2018 – May 2020*

- Troubleshoot POS systems and resolved technical issues
- Managed staff and ensured operational efficiency

## Relectromix LLC — Bellevue, WA

*Technician | March 2016 – April 2017*

- Repaired PC/Mac systems and configured Windows Server environments
- Performed diagnostics and customer-facing technical support

# Education

## North Idaho College — Current Student

C++, Python, Robotics, and Computer Science coursework (2 years)

## Code Fellows, Seattle

Full-Stack JavaScript Development Program (2017–2018)

## Bellevue College

Full-Stack JavaScript Coursework (2015–2017)

## Eastlake High School

Graduate (2011)

# References

Available upon request.

Academic reference: **Professor Ekaterina Miller**, North Idaho College.